St. George's University
Grenada, West Indies

BPM2 DM Small Group

HPWP 04: Selecting & Appraising Data for Self-Directed Learning

Student Instructions:

Before SDL SG

- Come to session prepared by reading the associated handout and instructions given below.

During SDL SG

- Students should listen respectfully to each other with an open mind and disagree respectfully
- Respectful and professional discussion during SG is *expected*
- When speaking or listening, be attentive to distinctions between fact, opinion, and perspective regarding what is said.
- Students should engage in discussion and utilize the notes, tools and technologies available to them within the session.
- Students are advised to use laptops/tablets to look up answers to the questions within the session.
- Self-directed learning (SDL) SGs are designed for students to advance their research skills by observing and practicing in real time
- Where you see the ">>" sign that means you will be required to do an activity e.g. look up information, design a research question.

Learning Objectives:

SOM.PB.3.BPM2.HPWP.045	Explain self-directed learning (SDL) by defining its characteristics, benefits and challenges.
SOM.PB.3.BPM2.HPWP.046	Apply the PICO framework to create appropriate clinical questions for a chosen scenario
SOM.PB.3.BPM2.HPWP.047	Develop search strategies by identifying key concepts, selecting appropriate databases, and refining searches to retrieve high-quality information.
SOM.PB.3.BPM2.HPWP.048	Appraise research articles by assessing their reliability, relevance to the question posed, and design quality and justify whether the evidence provided is appropriate for addressing complex clinical questions.
SOM.PB.3.BPM2.HPWP.049	Describe the key tenets of evidence-based medicine (EBM) that inform decision-making.

Questions:

1. Reflect on your engagement with self-directed learning (SDL) during medical school so far. Which strategies, resources, or approaches did you use that would count as SDL?

2. Reflect on your understanding of evidence-based medicine (EBM). What gaps or areas for further development have you identified in your knowledge or application of these concepts in SG cases?

3. As part of a study on judgment, over 200 physicians answered the following question:

   You are seeing a 78-year-old woman with a right upper lobe pulmonary nodule that was incidentally discovered on a chest CT after a motor vehicle accident. She is a 50 packyears smoker with anatomical emphysema on CT. The nodule is solid, solitary, 2.6 cm in diameter and spiculated; there is no thoracic adenopathy. She does not have a personal or family history of malignancy. The nodule is accessible by transthoracic needle biopsy. Based on this information, please estimate the following probability: Question: What is the overall probability that (1) the patient has malignancy and (2) a technically successful transthoracic needle biopsy will reveal malignancy on pathological examination?

   What is your answer?

4. Discuss your answer with your facilitator. Did you get it correct?
   If not, can you reflect on how you went wrong?
   If you did, how would you explain it to those with difficulty understanding it?

5. >> Construct a focused clinical question relevant to a patient encounter (or family member inquiry) using the PICO framework.

6. >> Apply the shortened Fresno rubric (from the handout) to search for and evaluate evidence that addresses your PICO clinical question.

7. Critically assess the validity (i.e. accuracy) of the research article(s) you have selected.
   Is the study's Methods section sound?
   What factors could influence the study's reliability (i.e. production of consistent results)?

8. Critically assess the Results section within your selected research paper in comparison to the example cited below.

   >> Locate the full text article by *Pfutzner et al., (2022): Chronic Uptake of A Probiotic Nutritional Supplement (AB001) Inhibits Absorption of Ethylalcohol in the Intestine Tract – Results from a Randomized Double-blind Crossover Study.*

   What is your interpretation of the probiotic findings? Justify your answer.
   What implications does this have for the study's reliability and applicability?

9. Critically assess the Results reported in the example cited below. In the example study, different methods of treating renal calculi were compared in order to discover which was more cost effective and successful. Of 1052 patients with renal calculi, 350 underwent open surgery, 350 percutaneous nephrolithotomy (PN), 328 extracorporeal shockwave lithotripsy (ESWL), and 24 both PN and ESWL. Treatment was defined as successful if stones were eliminated or reduced to less than 2 mm after three months.

Which approach is more successful? Justify your answer.
What other questions would you want answered before you could confidently make a conclusion about the findings?

>> Results provided below, but more study details can be found through *Charig et al., 1986. "Comparison of treatment of renal calculi by open surgery, percutaneous nephrolithotomy, and extracorporeal shockwave lithotripsy."*

TABLE II—*Success rate of treatment* (figures are numbers (%) of patients)*

	Group 1	Group 2	Overall
All open procedures	81 (93)	192 (73)	273 (78)
Percutaneous nephrolithotomy†	234 (87)	55 (69)	289 (83)

*Success defined as no stones at three months or stone reduced to particles <2 mm in size.
†52 with electrohydraulic lithotripsy, 69 with ultrasound.

>> Results provided above, but more study details can be found through *Charig et al., 1986: "Comparison of treatment of renal calculi by open surgery, percutaneous nephrolithotomy, and extracorporeal shockwave lithotripsy."*

10. Critically assess whether the Conclusions are generalizable from the example cited below.

>> Locate the full text article by *Rose et al., (2010): "Mood Food: Chocolate and Depressive Symptoms in a Cross-Sectional Analysis."*

Are the conclusions warranted by the Methods and Results (e.g. Figure 1)?
How applicable are these conclusions to wider clinical practice?

11.

You are part of a multidisciplinary healthcare team managing a 65-year-old patient with stable ischemic heart disease, multiple cardiovascular risk factors and angiography confirms moderate coronary artery disease. As you look into a treatment plan, you encounter conflicting recommendations in guidelines. Specifically, the American College of Cardiology (ACC), American

Heart Association (AHA), and Society for Cardiovascular Angiography and Interventions (SCAI) disagree with the American Association for Thoracic Surgery (AATS) and The Society of Thoracic Surgeons (STS).

What resources would you use to deepen your understanding of current coronary revascularization recommendations?
What strategies could you use to navigate conflicting evidence in clinical practice?
What role do patient preferences and multidisciplinary collaboration play in evidence-based decisions under uncertainty?

12. Final question - Take your time and answer the following:

It takes 5 machines 5 minutes to make 5 widgets. How long would it take 100 machines to make 100 widgets?

A. 100 minutes
B. 20 minutes
C. 5 minutes
D. 1 minute

Discuss the answer with your facilitator. Did you get it correct?

If not, can you reflect on how you went wrong?
If you did, how would you explain your approach to those who did otherwise?